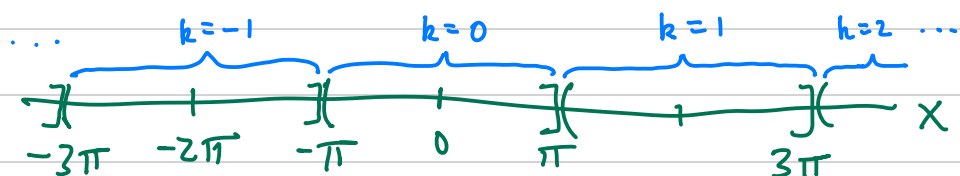


plan: periodic extensions, pointwise convergence examples
 shifting the interval, C_p^r functions
 decay of Fourier modes, derivative of a Fourier series
 uniform convergence, absolutely convergent series.

def: Suppose $I = [a, b)$ or $I = (a, b]$, with $-\infty < a < b < \infty$.

The periodic extension of a function $f: I \rightarrow \mathbb{C}$ (or $\mathbb{R}$)
 is defined for $x \in \mathbb{R}$ by $f_p(x) = f(x - kL)$, $L = b - a$,
 where for each x , $k \in \mathbb{Z}$ is chosen so $x - kL \in I$

picture for $(-\pi, \pi]$: ($L = 2\pi$)



existence/uniqueness of k : e.g. for $I = (a, b]$, we want

$$a < x - kL \leq b$$

$$x - b \leq kL < x - a$$

$$\frac{x-b}{L} \leq k < \frac{x-a}{L} = \frac{x-b+L}{L} = \frac{x-b}{L} + 1$$

$$k = \left\lceil \frac{x-b}{L} \right\rceil \text{ ceiling function.}$$

Every interval $[y, y+1)$ contains exactly one integer.

last time: if $f(x)$ is p.w. continuous on $[-\pi, \pi]$ and $f'(x_0^\pm)$ exist, then $S_N(x_0) \rightarrow \frac{f(x_0^+) + f(x_0^-)}{2}$ as $N \rightarrow \infty$

example: expand $f(x) = x$ in a Fourier series over $(-\pi, \pi]$.

plot $f_p(x)$ and $F(x) = \lim_{N \rightarrow \infty} S_N(x)$ over $[-3.5\pi, 3.5\pi]$

Repeat over $[0, 2\pi)$, making the plots over $[-4.5\pi, 4.5\pi]$

solution: $F(x) = \sum_{k=-\infty}^{\infty} C_k e^{ikh}$, $C_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-ikh} dx$

$$C_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = 0$$

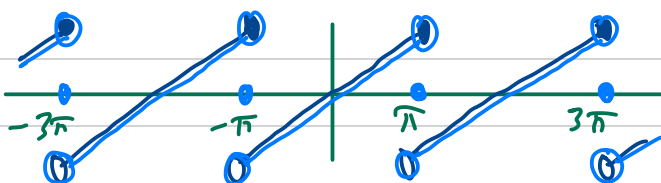
$$C_k = \frac{1}{2\pi} \left(\left. \frac{-x}{ik} e^{-ikh} \right|_{-\pi}^{\pi} + \left. \frac{1}{k^2} e^{-ikh} \right|_{-\pi}^{\pi} \right)$$

$$\begin{array}{l} x \\ | \\ 0 \end{array} \begin{array}{l} \nearrow + \\ \searrow - \end{array} \begin{array}{l} e^{-ikh} \\ \frac{-1}{ik} e^{-ikh} \\ \frac{-1}{k^2} e^{-ikh} \end{array}$$

$$= \frac{-1}{2ik} e^{-ikh\pi} - \frac{1}{2ik} e^{ikh\pi} + \frac{1}{2\pi k^2} (e^{-ikh\pi} - e^{ikh\pi})$$

$$e^{\pm i k \pi} = (e^{\pm i \pi})^k = (-1)^k, \quad \frac{1}{i} = -i$$

result: $C_k = \begin{cases} 0 & k=0 \\ \frac{(-1)^k}{k} i & k \neq 0 \end{cases}$



$f_p(x)$

$F(x)$

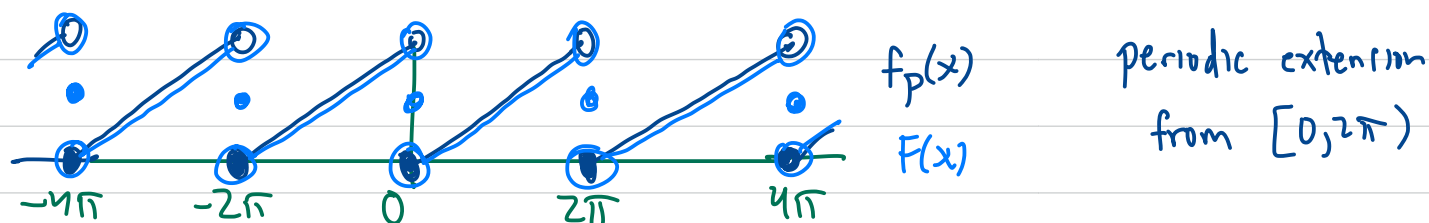
periodic extension
from $(-\pi, \pi]$

part 2: $c_k = \begin{cases} \frac{1}{2\pi} \int_0^{2\pi} x e^{-ikx} dx = \frac{i}{k} & (k \neq 0) \\ \frac{1}{2\pi} \int_0^{2\pi} x dx = \pi & (k = 0) \end{cases}$

x
 1
 0

$+$
 $-$
 $-$

e^{-ikh}
 e^{-ikh}
 e^{-ikh}

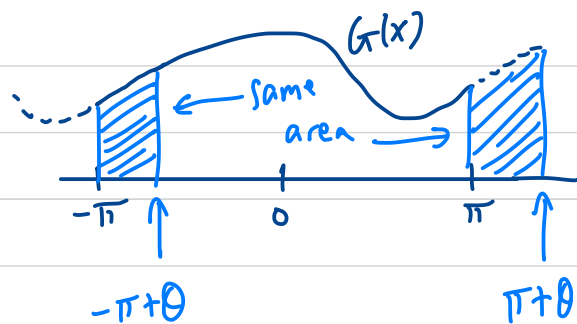


(Same $F(x)$ if we expand $f(x)$ over $[0, 2\pi]$. We only need the half-open interval $[0, 2\pi)$ to make $f_p(x)$ unambiguous.)

If $f(x)$ is 2π -periodic to begin with, then you get the same Fourier series by expanding over any period.

Reason: if $G(x)$ is 2π -periodic, then

$$\int_{-\pi+\theta}^{\pi+\theta} G(x) dx = \int_{-\pi}^{\pi} G(x) dx$$



$$c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} \underbrace{f(x)}_{G(x)} e^{-ikx} dx = \int_{-\pi+\theta}^{\pi+\theta} f(x) e^{-ikx} dx$$

same coefficients c_k , $F(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$ is valid over both $[-\pi, \pi]$ and $[-\pi+\theta, \pi+\theta]$

most common choices: $\theta = 0$, $\theta = \pi$

" " intervals: $[-\pi, \pi]$, $[0, 2\pi]$

(we usually use a closed interval in the continuous, periodic case)

Warning: here we translated the interval $[a, b]$. If we translate the function instead, the Fourier coefficients change by a phase factor.

$$g(x) = f(x-\theta) = \sum_k c_k e^{ik(x-\theta)} = \sum_k \underbrace{c_k e^{-ik\theta}}_{d_k} e^{ikx}$$

alternative derivation:

$$d_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-\theta) e^{-ikx} dx = \frac{1}{2\pi} \int_{-\pi-\theta}^{\pi-\theta} \underbrace{f(\xi) e^{-ik\xi}}_{f(\xi) \text{ is periodic}} e^{-ik\theta} d\xi$$

$\xi = x - \theta, d\xi = dx$

$$= \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(\xi) e^{-ik\xi} d\xi \right) e^{-ik\theta} = c_k e^{-ik\theta}$$

def: $f \in C^r(a, b)$ if $\begin{cases} f^{(j)}(x) \text{ exists and is continuous on } (a, b) \\ \text{for } 0 \leq j \leq r. \end{cases}$

def: $f \in C^r[a, b]$ if $\begin{cases} f \in C[a, b] & (\text{continuous even at the endpoints}) \\ f \in C^r(a, b) \\ f^{(j)}(x) \text{ extends continuously to } [a, b] \text{ for } 1 \leq j \leq r \end{cases}$

(this just means $\lim_{x \rightarrow a^+} f^{(j)}(x)$, $\lim_{x \rightarrow b^-} f^{(j)}(x)$ exist)

one can show that if $f \in C^r[a, b]$, the one-sided derivatives

$$f^{(j)}(a^+) = \lim_{h \rightarrow 0^+} \frac{f^{(j-1)}(a+h) - f^{(j-1)}(a^+)}{h}$$

$$\text{and } f^{(j)}(b^-) = \lim_{h \rightarrow 0^-} \frac{f^{(j-1)}(b+h) - f^{(j-1)}(b^-)}{h}$$

$$\text{also exist, and } f^{(j)}(a^+) = \lim_{x \rightarrow a^+} f^{(j)}(x)$$

$$f^{(j)}(b^-) = \lim_{x \rightarrow b^-} f^{(j)}(x)$$

here's a proof, focusing on the left endpoint.

$$\text{define } A_j = \lim_{x \rightarrow a^+} f^{(j)}(x), \quad 0 \leq j \leq r$$

Fix $1 \leq j \leq r$ and define

$$g_j(x) = \begin{cases} A_{j-1} & x = a \\ f^{(j-1)}(x) & x \in (a, b) \end{cases}$$

Let $\varepsilon > 0$ and choose $\delta > 0$ small enough that

$$|f^{(j)}(x) - A_j| < \varepsilon \text{ for } a < x < \delta.$$

Then for $0 < h < \delta$ we have

$$\left| \frac{g_j(a+h) - g_j(a)}{h} - A_j \right| = \left| g'_j(\xi) - A_j \right| = \left| f^{(j)}(\xi) - A_j \right| < \varepsilon$$

by the mean value theorem,
such a $\xi \in (a, a+h)$ exists.

So the limit $f^{(j)}(a^+) = \lim_{h \rightarrow 0^+} \frac{f^{(j-1)}(a+h) - A_{j-1}}{h}$

exists and equals $A_j = \lim_{x \rightarrow a^+} f^{(j)}(x)$ for $1 \leq j \leq r$

The first time ($j=1$), $A_0 = f(a) = f^{(0)}(a^+)$

and after that, we have already established that

$A_{j-1} = f^{(j-1)}(a^+)$ from the previous iteration.

def: $f \in C_p^r[a, b]$ if $f \in C^r[a, b]$ and $f^{(j)}(a^+) = f^{(j)}(b^-)$ for $0 \leq j \leq r$.

↗ we often write $C_p[a, b]$ for $C_p^0[a, b]$. $p \leftrightarrow$ "periodic"

equivalently, the periodic extension $f_p(x)$ from $[a, b)$ belongs to $C^r(\mathbb{R})$

def:

f is piecewise C^r on $[a, b]$ if $\exists$ points x_m with $\swarrow$ the "knots"

$$a = x_0 < x_1 < \dots < x_\mu = b$$

such that

$$f \in C^r(x_{m-1}, x_m) \quad 1 \leq m \leq \mu$$

and

$$\left(\lim_{x \rightarrow x_{m-1}^+} f^{(j)}(x) \right), \left(\lim_{x \rightarrow x_m^-} f^{(j)}(x) \right) \text{ exist for } \left(\begin{array}{l} 0 \leq j \leq r \\ 1 \leq m \leq \mu \end{array} \right).$$

We don't assume $f^{(j)}(x_m^-) = f^{(j)}(x_m^+)$ or $f^{(j)}(a^+) = f^{(j)}(b^-)$ for any j or m , including $j=0$, and $f(x_m)$ can be assigned any value for $0 \leq m \leq \mu$

Corollary of main theorem of Lec 10:

If f is p.w.- C^1 on $(-\pi, \pi]$, then $f_p(x)$ is p.w.- C^1 on $\mathbb{R}$

and $F(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$ satisfies $F(x) = \frac{1}{2}(f_p(x^-) + f_p(x^+))$ for all $x \in \mathbb{R}$.

$$\text{In particular, } F(\pi) = \frac{1}{2} \left(\underbrace{f(\pi^-)}_{f_p(\pi^-)} + \underbrace{f(-\pi^+)}_{f_p(\pi^+)} \right)$$

Decay of Fourier coefficients (smooth functions $\Leftrightarrow$ faster decay)
(very important in practice!)

Theorem: If f is p.w. C^1 on $[-\pi, \pi]$ then $\exists M_1 > 0$ s.t.

$$|C_k| \leq \frac{M_1}{|k|} \quad (k \neq 0)$$

(Remark: This is a slow decay rate, and doesn't imply that the Fourier series converges uniformly, but it's an important first step.)

proof: if $k \neq 0$, we have

$$C_k = \frac{1}{2\pi} \sum_{m=1}^M \int_{x_{m-1}}^{x_m} f(x) e^{-ikx} dx$$

$$\left\{ \begin{array}{l} u = f(x) \\ du = f'(x) dx \\ dv = e^{-ikx} dx \\ v = \frac{1}{-ik} e^{-ikx} \end{array} \right.$$

$$= \frac{1}{2\pi} \sum_{m=1}^M \left[\underbrace{\frac{f(x)}{-ik} e^{-ikx}}_{\substack{f(x_m^-) e^{-ikx_m} - f(x_{m-1}^+) e^{-ikx_{m-1}}}} \Big|_{x_{m-1}}^{x_m} + \int_{x_{m-1}}^{x_m} \frac{f'(x)}{ik} e^{-ikx} dx \right]$$

(*)

Since $|e^{ikx}| = 1$, the triangle inequality gives $|C_k| \leq \frac{M_1}{|k|}$

$$\text{with } M_1 = \frac{1}{2\pi} \sum_{m=1}^M \left[|f(x_m^-)| + |f(x_{m-1}^+)| + \int_{x_{m-1}}^{x_m} |f'(x)| dx \right]$$

We didn't assume f is continuous to get $|c_k| \leq \frac{M}{|k|}$.

If $f \in C_p[-\pi, \pi]$ and f is p.w. C^1 then all the boundary terms in $(*)$ cancel due to $f(x_m^+) = f(x_m^-)$ and $f(a^+) = f(b^-)$ and we have

$$(**) \quad c_k = \frac{1}{ik} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) e^{-ikx} dx \right) \quad (k \neq 0)$$

Also note that when $k=0$, $\frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) dx = \frac{f(\pi) - f(-\pi)}{2\pi} = 0$

$$\text{so } f'(x) = \sum_{k=-\infty}^{\infty} \underbrace{ikc_k}_{\text{the Fourier coefficients of } f'(x)} e^{ikx}$$

So one can differentiate the Fourier series term by term to get the expansion for $f'(x)$ as long as $f \in C_p[-\pi, \pi]$ and f is p.w. C^1 on $[-\pi, \pi]$.

The decay rate improves if f is p.w. C^2 :

Theorem: if $f \in C_p[-\pi, \pi]$ and f is p.w. C^2 on $[-\pi, \pi]$ then $|c_k| \leq \frac{M_2}{|k|^2}$

proof: rewrite $(**)$ as follows

$$c_k = \frac{1}{2\pi} \sum_{m=1}^M \int_{x_{m-1}}^{x_m} \frac{f'(x)}{ik} e^{-ikx} dx$$

and integrate by parts again:

$$c_k = \left(\frac{1}{ik}\right)^2 \frac{1}{2\pi} \sum_{m=1}^M \left[-f'(x) e^{-ikx} \right]_{x_{m-1}}^{x_m} + \int_{x_{m-1}}^{x_m} f''(x) e^{-ikx} dx$$

(these cancel again if $f \in C_p^1[-\pi, \pi]$)

$$\therefore |c_k| \leq \frac{M_2}{|k|^2}$$

otherwise they show up in M_2

with

$$M_2 = \frac{1}{2\pi} \sum_{m=1}^M \left[|f'(x_m^-)| + |f'(x_{m-1}^+)| + \int_{x_{m-1}}^{x_m} |f''(x)| dx \right]$$

This argument can be repeated until the boundary terms fail to cancel, or until f runs out of derivatives:

Theorem: If $r \geq 2$ and $\begin{cases} f \in C_p^{r-2}[-\pi, \pi] \text{ and} \\ f \text{ is p.v. } C^r \text{ on } [-\pi, \pi] \end{cases}$

$$\text{then } \exists M_r \text{ s.t. } |c_k| \leq \frac{M_r}{|k|^r} \quad (k \neq 0)$$

Corollary: If $r \geq 2$, $f \in C_p^{r-2}[-\pi, \pi]$ and f is p.v. C^r on $[-\pi, \pi]$
 then $P_N f \rightarrow f$ uniformly.

Proof: We already know $P_N f \rightarrow f$ pointwise since $f \in C_p[-\pi, \pi]$.

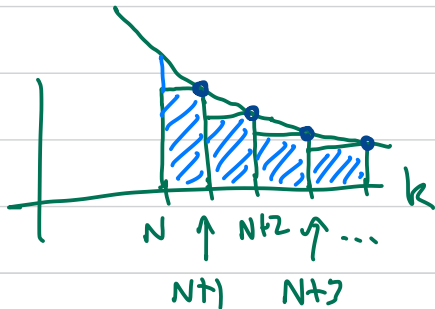
$$P_N f(x) = S_N(x) = \sum_{k=-N}^N c_k e^{ikx}$$

$$|P_N f(x) - f(x)| = \left| \sum_{k=N+1}^{\infty} (c_k e^{ikx} + c_{-k} e^{-ikx}) \right|$$

still a
pointwise
formula

$$\leq \sum_{N+1}^{\infty} (|c_k| + |c_{-k}|) \leq \sum_{N+1}^{\infty} \frac{2M}{k^2}$$

$M = M_2$



$$\leq 2M \int_N^{\infty} \frac{1}{k^2} dk = 2M \left(\frac{-1}{k} \right)_N^{\infty} = \frac{2M}{N}$$

Let $\varepsilon > 0$ and choose any $N_0 > \frac{2M}{\varepsilon}$.

Then for $N \geq N_0$,

$$\|P_N f - f\|_{\infty} = \max_{-\pi \leq x \leq \pi} |P_N f(x) - f(x)| \leq \frac{2M}{N} \leq \frac{2M}{N_0} < \varepsilon$$

$\therefore P_N f \rightarrow f$ uniformly.

Comments on infinite series. Let V be a complete normed space (it could just be $\mathbb{R}$ or $\mathbb{C}$).

saying that a series $\sum_{k=1}^{\infty} a_k$ with $a_k \in V$

converges means that the sequence of partial

sums $s_n = \sum_{k=1}^n a_k$ converges as $n \rightarrow \infty$.

Since V is complete, this is equivalent to $\{s_n\}$

being a Cauchy sequence: for any $\varepsilon > 0 \exists N$

s.t. for $m, n \geq N$ then $\|s_n - s_m\| < \varepsilon$.

A sufficient condition for $\sum_{k=1}^{\infty} a_k$ to converge

is that $\sum_{k=1}^{\infty} \|a_k\|$ converges in $\mathbb{R}$. (The sum is finite)

Suppose so and let $\varepsilon > 0$. Pick N large enough that

if $n > m \geq N$ then $\sum_{k=m}^{n-1} \|a_k\| < \varepsilon$.

But then

$$\|s_n - s_m\| = \left\| \sum_{k=m}^{n-1} a_k \right\| \leq \sum_{k=m}^{n-1} \|a_k\| < \varepsilon$$

so $\{s_n\}$ is a Cauchy sequence and $\sum_{k=1}^{\infty} a_k$ converges.

Series for which $\sum_{k=1}^{\infty} \|a_k\| < \infty$ are called absolutely convergent.